

MBR Intelligent Site Assistant

User Guide · Version 0.6.0

A self-hosted conversational site search for WordPress.
No external APIs. No monthly fees. Nothing leaves your server.

Published by Little Web Shack · littlewebshack.com

Author: Robert Palmer

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1. Introduction

The MBR Intelligent Site Assistant is a WordPress plugin that adds a conversational chat widget to your site. Visitors type questions in plain English, and the assistant responds with short messages and links to the most relevant pages on your site.

The plugin runs entirely on your own server. There is no external API call for each query, no monthly subscription, and no visitor data sent to a third party. The ranking is done in pure PHP and MySQL using a well-understood information retrieval algorithm (BM25), layered with intent matching for common questions and synonym expansion for natural phrasing.

1.1 What it does

- Adds a floating chat bubble, or an inline widget via shortcode.
- Indexes your published posts and pages automatically when they are saved.
- Matches common questions ("contact", "pricing") to pre-written answers, with HTML support for links and basic formatting.
- Expands queries with synonyms so "WP" finds content tagged "WordPress".
- Logs queries and thumbs-up / thumbs-down feedback so you can tune responses.
- Provides full admin UIs for intents, synonyms, and widget appearance — no code editing required.
- Five colour-coordinated themes plus an optional glassmorphism effect, all configurable from the admin.

1.2 What it is not

Despite the word *assistant*, this plugin is not a large language model and does not generate text. It retrieves and surfaces content you have already written, and answers a configurable list of canned questions. If a visitor asks something you have not written about, the assistant will politely say it could not find anything, and invite them to rephrase.

This is a deliberate design choice. The plugin produces predictable, auditable responses with no hallucination risk — and it does it without sending your visitors' questions to an outside service.

2. What's new since version 0.3.2

If you are upgrading from v0.3.2 (the previous published version of this guide), here is the headline summary. Each item is covered in detail in its own chapter.

2.1 Version 0.4.0 — intents admin UI

- New **Tools** → **MBR ISA Intents** page for adding, editing, and deleting intents without touching code.
- Per-intent **disabled** toggle so you can keep an intent in the list without firing it — useful for A/B testing.
- Intent responses now support basic HTML (links, bold, line breaks, lists, etc.). Sanitised on save with `wp_kses_post()`.
- Inline test panel: type a query and see which intent (if any) would match before saving.

- Inline regex validation for `re:` triggers — bad patterns are rejected on save with the offending trigger named.

2.2 Version 0.5.x — synonyms admin UI

- New **Tools** → **MBR ISA Synonyms** page mirroring the intents UX.
- Test panel showing token expansion before and after synonym lookup, with stems labelled by their original word in brackets when the two differ (so `websit (website)` reads naturally).
- Both newline-separated and comma-separated input accepted — paste from a thesaurus and it just works.

2.3 Version 0.6.0 — appearance admin UI

- New **Tools** → **MBR ISA Appearance** page for choosing the chat-widget colour scheme.
- Five colour-coordinated presets: Mocha (default Catppuccin dark), Slate Light, Ocean, Sunset, and Forest.
- Glassmorphism toggle layers on top of any preset — translucent panel with a backdrop blur, designed for sites with rich page backgrounds.
- Live interactive preview rendered with the actual widget CSS over an animated gradient stage, so what you see is exactly what visitors will get.

2.4 Diagnostic page improvements

The diagnostic page at **Tools** → **MBR ISA Diagnostic** now has three quick-access buttons at the top linking to the new admin pages, so the workflow is a single click away from any diagnostic action.

3. How it works

When a visitor submits a question, the plugin runs it through a five-stage pipeline. Understanding this pipeline will help you tune the plugin effectively.

3.1 The query pipeline

- **Intent match.** The raw query is checked against configured intent triggers. If any trigger matches as a substring (or regex, if prefixed with `re :`), the matching intent's canned response is returned immediately. Search is skipped. Disabled intents are ignored.
- **Tokenisation.** If no intent matched, the query is lowercased, stripped of punctuation, split on whitespace, filtered against an English stopword list, and each surviving word is reduced to its stem using a Porter stemmer.
- **Synonym expansion.** Each stemmed token is looked up in the synonym map. Matching terms add their siblings to the token list.
- **Ranking.** The expanded tokens are scored against the inverted index using BM25, with title matches weighted higher than body matches.
- **Formatting.** Results are classified by confidence (high, medium, low, or none), a matching message is chosen, and short highlighted snippets are built around the query terms.

3.2 Confidence levels

The responder picks one of four confidence levels based on the top result's BM25 score and how much it dominates the second result. Each level changes both the framing message and how many results are returned.

Level	Rule	Results	Typical framing
High	Top score ≥ 1.5 and $\geq 1.5\times$ the runner-up	1	"Here is what I found."
Medium	Top score ≥ 1.0	Up to 3	"These look relevant."
Low	Top score > 0 but below 1.0	Up to 3	"I'm not sure, but these might help."
None	No results	0	"I couldn't find anything."

3.3 The four database tables

The plugin creates four custom tables, all prefixed with your WordPress table prefix followed by `mbrisa_`:

Table	Purpose
<code>mbrisa_terms</code>	Dictionary of every unique token, with document frequency for BM25 IDF.
<code>mbrisa_documents</code>	One row per indexed post or page: title, URL, token count, content hash, and a short excerpt.

Table	Purpose
mbrisa_postings	Inverted index — which terms appear in which documents, with frequency per field (title / content / excerpt).
mbrisa_queries	Log of every query asked through the widget, with intent match, result count, top score, and feedback.

4. Installation

4.1 Requirements

- WordPress 5.8 or later
- PHP 7.4 or later
- Standard MySQL / MariaDB — no special extensions required
- No external services, API keys, or network access

4.2 Installing the plugin

Because this plugin is distributed through Little Web Shack rather than the WordPress.org repository, installation is a manual upload:

- Download `mbr-intelligent-site-assistant.zip` from littlewebshack.com.
- In your WordPress admin, go to **Plugins** → **Add New**, then click **Upload Plugin**.
- Choose the ZIP file and click **Install Now**, then **Activate Plugin**.
- On activation, the four database tables are created and the default settings are seeded. Any existing published posts and pages will not be indexed automatically at this point — you need to run a full reindex once (see § 5).

Note. The floating chat widget is off by default on a fresh install. This is intentional — the plugin wants you to run a reindex and check the tokeniser before the widget starts talking to real visitors.

5. First-time setup

After activation, open **Tools** → **MBR ISA Diagnostic** in your WordPress admin. The diagnostic page is the central hub — from there you can reindex, test queries, configure the widget, and reach the three specialised admin pages (intent, synonyms, appearance) via quick-access buttons at the top. The typical first-run sequence is: reindex, test, configure widget, pick a theme, enable the widget.

5.1 Run a full reindex

Scroll to the Index Status section and click **Run Full Reindex**. This wipes the index tables and rebuilds them from every published post and page (or whichever post types you have enabled in the settings). Indexing 100 posts typically takes a few seconds on a modest host.

When it finishes you should see non-zero values for Documents indexed, Unique terms in dictionary, and Total postings.

5.2 Test with the chat tester

At the bottom of the diagnostic page, the Chat Tester runs the full query pipeline exactly as the public widget would. Type a question you expect the plugin to handle — for example, *"how do I contact you"* (which should match the built-in Contact intent), or a word you know appears on several pages of your site.

Expand the *Raw response payload* details to see the exact JSON the widget will receive. This is invaluable when diagnosing why a query is not producing the result you expected.

5.3 Configure and enable the widget

Scroll up to **Widget Settings**. The important fields are:

Setting	Purpose
Floating widget	Tick to show the chat bubble on every page. Leave off if you only want to embed the widget inline via shortcode.
Position	Bottom-right (default) or bottom-left.
Widget title	Header text in the chat panel. Default: "Site Assistant".
Greeting message	First message the assistant shows when the panel opens.
Input placeholder	Ghost text in the input box.

Save your changes. If you ticked **Floating widget**, the bubble will now appear on the front end of your site.

5.4 Pick a theme (optional)

Click the **Appearance** → button at the top of the diagnostic page. The default theme is Mocha (Catppuccin dark purple), which is what previous versions of the plugin shipped. If a different palette suits your site better, choose one of the four other presets — the live preview shows what the widget will look like, and the glassmorphism toggle adds an optional translucent effect. See § 10 for full details.

6. The diagnostic dashboard

The Diagnostic page at **Tools** → **MBR ISA Diagnostic** is the operational hub of the plugin. Three quick-access buttons at the top take you to the specialised admin pages; the body of the diagnostic page contains everything else.

6.1 Quick-access buttons

At the top of the diagnostic page there are three buttons:

- **Manage intents** → — opens the intents admin page (§ 8).
- **Manage synonyms** → — opens the synonyms admin page (§ 9).
- **Appearance** → — opens the theme and effects admin page (§ 10).

6.2 Plugin Status

Quick sanity check: plugin version, database schema version, server PHP version, and the REST endpoint URL the widget will post to. If you are troubleshooting a broken widget, confirm this URL returns a valid response when you POST to it.

6.3 Database Tables

A four-row table showing whether each of the plugin's custom tables exists. Under normal operation all four should show a green tick. If any table is missing, deactivate and reactivate the plugin to trigger the activation routine again.

6.4 Index Status

Shows document, term, and posting counts, plus the timestamp of the last full reindex. Also hosts the **Run Full Reindex** button. A reindex is safe to run at any time — it wipes and rebuilds the index tables from scratch. You should run one after:

- Bulk-importing or bulk-editing posts
- Changing the list of enabled post types
- Modifying the stopword list or tokeniser (unlikely in normal use)
- Restoring from a backup

6.5 Widget Settings

All of the visible widget configuration lives here. See § 5.3 for a field-by-field breakdown and § 7 for more on widget placement.

6.6 Feedback & Queries

Seven-day statistics — total queries, thumbs up, thumbs down, how many returned zero results, and how many hit an intent. Below that is a table of the 30 most recent queries. Rows tinted red are zero-result queries or queries that received a thumbs-down. These are the ones worth looking at first, because each one represents either a missing intent, a missing synonym, or a gap in your site's content.

6.7 Tokeniser Tester

Paste any text and see exactly what the tokeniser does to it: original, cleaned, split, after stopwords, after stemming. Mostly useful when debugging — for instance, if you are surprised that a particular word is not matching, tokenise the query and tokenise the page title and compare.

6.8 Search Tester (raw BM25)

Runs the search pipeline without intent matching or synonym expansion, and shows a full trace of which terms were looked up and their scores. Useful when you want to confirm the ranking itself is behaving correctly, independent of the higher-level pipeline stages.

6.9 Chat Tester (full pipeline)

Runs exactly what the public widget runs, and shows the conversational output with the confidence badge and the raw JSON payload. This is the primary tool for tuning intents and synonyms.

7. The widget: placement and customisation

7.1 Floating bubble mode

With **Floating widget** ticked in settings, a small chat bubble appears on every page of the front end, pinned to the bottom-right or bottom-left. Clicking the bubble opens the chat panel. The widget is automatically skipped on feeds, embeds, the favicon endpoint, and the robots.txt endpoint.

7.2 Inline mode via shortcode

To embed the widget within the flow of a specific page — for example, a dedicated "Ask" page — use the shortcode:

```
[mbr_isa_chat]
```

The shortcode supports four optional attributes that override the global settings for that one embed:

Attribute	Effect
title	Overrides the widget title shown in the panel header.
greeting	Overrides the first assistant message.
placeholder	Overrides the input placeholder text.
height	Sets the panel height — any valid CSS unit, e.g. 500px, 80vh.

Example:

```
[mbr_isa_chat title="Ask us anything" greeting="What are you looking for?" height="600px"]
```

7.3 Running both at once

It is safe to leave the floating widget on and also place a shortcode on a specific page. Each instance gets its own unique ID and operates independently. Session storage is shared, so a conversation started in one instance persists if the visitor switches to the other.

7.4 Assets and conditional loading

The widget's CSS and JavaScript are only enqueued on pages where they are actually needed: either because the floating widget is enabled site-wide, or because the current page contains the shortcode. Pages that use neither do not load the assets.

7.5 Theming

Every visual aspect of the widget is themeable from the Appearance admin page (§ 10). Both modes — floating and inline — pick up whichever preset and effects you have configured site-wide. There is no shortcode override for theme; it is intentionally a single setting per site.

8. Intents — managing through the admin UI

An intent pairs a list of trigger phrases with a canned response. When a query matches any trigger, the intent's response is returned immediately — no search, no ranking. Intents are the single biggest usability win in a site assistant, because they let you give a perfect answer to every common question.

From version 0.4.0 onwards, intents are managed through a dedicated admin page rather than by editing the option in code. Find it at **Tools** → **MBR ISA Intents**, or via the **Manage intents** → button on the diagnostic page.

8.1 The intents page layout

The page is divided into four sections, top to bottom:

- **Test panel** — type a query and see which intent (if any) would match, with the response, label, and confidence.
- **Existing intents** — one card per intent, each containing its own form.
- **Add a new intent** — a blank card at the bottom for creating intents from scratch.
- **Reset to defaults** — replaces all intents with the shipped defaults (with a JS confirmation).

8.2 Built-in intents

The plugin ships with five default intents, used as a starting point on first install. Once you save any change, the option becomes your own and the defaults stop applying — but you can always reset to them via the button at the bottom of the page.

ID	Example triggers
contact	contact, email address, get in touch, how do I reach
pricing	how much, what do you charge, pricing, your rates, cost of
services	what services, what do you do, what do you offer, can you build
hours	opening hours, what hours, when are you open, business hours
help	need help, can you help, I am stuck, not working

8.3 Editing an intent

Each intent card has six fields:

- **ID** — lowercase, dashes, underscores. Used internally; appears in the diagnostic logs.
- **Label** — human-friendly name shown in test results and diagnostics.
- **Confidence** — a number between 0.0 and 1.0. When more than one intent matches, the higher confidence wins.
- **Triggers** — one trigger per line. Substring match by default; prefix with `re:` for regex.
- **Response** — the message shown to visitors. Basic HTML allowed (links, bold, italic, line breaks, lists). Sanitised on save.

- **Disabled** — when ticked, the intent is kept in the list but does not fire. Useful for A/B testing or temporarily silencing an intent.

Each card has its own **Save changes** button, so an in-progress edit to one intent can never lose another's edits. Each card also has a **Delete intent** button (with confirmation).

8.4 HTML in responses

Responses go through `wp_kses_post()` on save, which permits the same set of tags WordPress allows in post content — including links with `<a href>`, `<strong>`, `<em>`, `<br>`, paragraphs, and lists. Anything outside that whitelist is stripped at save time, so the stored response is always safe to render.

On the front end, intent responses are rendered with the HTML preserved. Search responses (which use a different code path) remain plain text, as they always have.

8.5 How matching works

- Triggers are matched case-insensitively as substrings of the raw query.
- A trigger prefixed with `re:` is treated as a regex (useful for word-boundary matches, e.g. `re:\bhours?\b`).
- If more than one intent matches, the one with the highest confidence value wins.
- Disabled intents are skipped during matching.
- Empty queries are never matched; neither are intents missing an ID, triggers, or a response.

8.6 Validation rules

When you save an intent, the admin enforces a few rules to prevent obvious mistakes:

- ID must be unique. Trying to save a new intent with the same ID as an existing one is rejected with a clear error.
- ID must consist only of lowercase letters, numbers, dashes, and underscores.
- At least one trigger is required.
- Each `re:` regex trigger must be a valid pattern. Invalid regexes are rejected on save with the offending trigger named.
- The response, after stripping HTML, cannot be empty.

Tip. Use the Recent queries list on the diagnostic page to find questions that returned zero results, then add triggers or intents to cover them.

9. Synonyms — managing through the admin UI

A synonym group is a list of terms that should be considered equivalent for searching. If any term in a group appears in a query, the others are added to the search before ranking. Matching happens on stems, not raw words, so "shops" and "shop" are already equivalent before synonyms kick in.

From version 0.5.0 onwards, synonym groups are managed through a dedicated admin page. Find it at **Tools** → **MBR ISA Synonyms**, or via the **Manage synonyms** → button on the diagnostic page.

9.1 Built-in groups

The defaults cover the common web-industry vocabulary. Each row is one group:

Group	Terms
1	wp, wordpress
2	site, website, webpage
3	ecommerce, e-commerce, shop, store, online shop
4	price, pricing, cost, fee, rate, quote
5	contact, email, reach, get in touch
6	help, support, assistance
7	plugin, extension, add-on, add-on
8	theme, template, skin
9	build, develop, create, make
10	seo, search engine optimisation, search engine optimization

9.2 Editing a group

Each group card has a single textarea. Enter one term per line — multi-word phrases like *"search engine optimisation"* are fine on a single line. Both newline-separated and comma-separated input is accepted on save, so pasting from a thesaurus also works.

A group needs at least two unique terms after parsing. Single-term groups do nothing useful, so they are rejected on save with a clear error.

Each card has its own Save and Delete buttons, mirroring the intents page. The header preview shows the first five terms of the group as a code chip — handy when scanning a long list.

9.3 The test expansion panel

Type a query into the test panel and the page shows two rows of token chips:

- **Original tokens** — the words from your query after stemming.

- **Added by synonyms** — the extra tokens that synonym expansion contributed, highlighted in green.

Each chip shows the stem followed by the original word in lighter brackets when the two differ. So `websit (website)` reads naturally, while `post (unchanged by stemming)` shows as just `post`. Visitors never see the stemmed form anywhere — this is admin-side feedback only.

Note. The Porter stemmer collapses related words to a common root so the index can match them as one. That's why "website" stems to "websit" and "running" stems to "run" — it's how the search actually works internally. The bracketed original word is purely for readability in the test panel.

9.4 When to add a synonym vs an intent

Add a synonym when the missing phrasing is something visitors say but your content uses a different word. "Delivery" and "shipping" might both refer to the same page. Synonyms make the search find what's already on your site.

Add an intent when the question has a canned answer that does not correspond to any single page on your site. "What are your opening hours?" might not have a dedicated page — but you still want a good answer.

10. Appearance — themes and glass effect

From version 0.6.0 onwards, the chat-widget colour scheme is configurable through a dedicated admin page. Find it at **Tools** → **MBR ISA Appearance**, or via the **Appearance** → button on the diagnostic page.

The page has three areas: a live preview at the top, a colour-preset selector in the middle, and an effects toggle at the bottom. Changes to the controls update the preview immediately; clicking **Save appearance** commits them to the front end.

10.1 The live preview

The preview renders an actual chat-widget structure using the same CSS as the public widget, over an animated gradient background that slowly cycles through a purple-pink-magenta palette. The moving background is intentional — the glass effect (see § 10.3) only really shows itself when there is something behind the panel for it to blur.

Clicking a preset card or toggling the glass checkbox updates the preview without saving. You can browse presets freely; nothing is committed until you press **Save appearance**.

10.2 The five presets

Each preset is a coordinated palette that overrides every colour in the widget. The presets are:

Preset	Kind	Description
Mocha	Dark	Catppuccin Mocha — the original default. Deep purple panel with mauve and lavender accents. Suits dark and design-led sites.
Slate Light	Light	Clean neutral greys with indigo accent. The corporate-friendly default — works on almost any site.
Ocean	Light	Soft light blues with teal accent. Reads as friendly and calm; good for service businesses, healthcare, education.
Sunset	Light	Warm peach base with coral accents. Distinctive and warm; suits hospitality, food, lifestyle sites.
Forest	Dark	Deep green forest with sage accent. The dark counterpart to Mocha for sites that want dark but not purple.

Existing installs upgrading from before v0.6.0 will see no visual change — Mocha is the default, and the original Catppuccin Mocha CSS is preserved exactly under that preset name.

10.3 Glassmorphism

The Effects section has a single toggle: **Glassmorphism**. When enabled, the chat panel becomes translucent with a backdrop blur, the bubble button gets a frosted-glass treatment, and the conversation bubbles within the panel get a softer glass overlay.

Glass works in combination with any of the five presets. The exact look differs by preset:

- On dark presets (Mocha, Forest), glass produces a smoked-glass effect — dark and translucent.

- On light presets (Slate Light, Ocean, Sunset), glass produces a frosted-glass effect — bright and translucent.

Glass relies on the CSS `backdrop-filter` property, which has wide modern browser support (Chrome, Edge, Firefox, Safari on current versions). Older browsers fall back to a partially transparent panel without the blur — readable, just less fancy.

Tip. Glass looks best over rich backgrounds. On a white-on-white page the effect is subtle by design — there is nothing visually interesting behind the panel for the blur to act on. On hero images, gradients, or photo-heavy pages, the effect carries.

10.4 Saving and applying

Clicking **Save appearance** writes both `theme_preset` and `theme_glass` into the existing `mbr_isa_settings` option, preserving every other widget setting (title, greeting, placeholder, position, etc.). The front-end widget picks up the new theme on the next page load — no cache flush required.

11. Indexing behaviour

11.1 What gets indexed

By default, published posts and pages are indexed. The indexer pulls the title, post content, and excerpt fields, tokenises them, and stores per-field term frequencies so that title matches can be weighted higher than body matches at query time.

To enable additional post types, update the `enabled_post_types` entry in the plugin settings option:

```
$s = get_option( 'mbr_isa_settings', [] );
$s['enabled_post_types'] = [ 'post', 'page', 'product' ];
update_option( 'mbr_isa_settings', $s );
// Then run a full reindex from the diagnostic page.
```

11.2 Automatic re-indexing on save

The indexer hooks `save_post`, `deleted_post`, and `trashed_post`. Every time you save or publish a post of an enabled type, the indexer recomputes the content hash; if it has changed, the document's entry in the index is updated. If the post is unpublished, trashed, or deleted, it is removed from the index. Revisions and autosaves are ignored.

11.3 Field weighting

BM25 is applied per field with the following default weights:

Field	Default weight
Title	3.0
Excerpt	1.5
Content	1.0

Weights are stored in the plugin settings option as `field_weight_title`, `field_weight_excerpt`, and `field_weight_body`. The BM25 parameters `bm25_k1` (default 1.2) and `bm25_b` (default 0.75) are also stored there.

12. REST API

The chat widget talks to the plugin via two REST endpoints, both under the `mbr-isa/v1` namespace. Both are public (no authentication) but nonce-protected when called from the widget, and rate-limited per IP.

Note. If you have a security or REST-hardening plugin installed (such as MBR WP Performance), make sure `mbr-isa/v1` is added to its REST allowlist. Otherwise the plugin will block the widget's requests and visitors will see a generic error message. See § 14 for the troubleshooting walk-through.

12.1 POST `/wp-json/mbr-isa/v1/ask`

Accepts a user question and returns the full structured chat response.

Request body:

```
{
  "query": "how do I contact you",
  "session_id": "optional-32-char-session-id"
}
```

Successful response (intent hit):

```
{
  "type": "intent",
  "intent_id": "contact",
  "intent_label": "Contact",
  "message": "You can get in touch via the contact form...",
  "message_html": "You can get in touch via the <a href=...>contact form</a>...",
  "message_format": "html",
  "results": [],
  "suggestions": ["Ask me something else about this site"],
  "session_id": "...",
  "query_id": 1234
}
```

From version 0.4.0, intent responses include `message_html` (the sanitised HTML response, rendered by the widget via `innerHTML`) and a `message_format` hint of `html`. The plain `message` field is the HTML-stripped version, kept as a safe fallback for any consumer that cannot render HTML. Search responses are unchanged — they remain plain text.

Successful response (search hit, medium confidence):

```
{
  "type": "search_results",
  "confidence": "medium",
  "message": "These look relevant:",
  "results": [
    { "title": "...", "url": "...", "snippet": "... <mark>match</mark> ...", "score": 2.14 }
  ],
  "suggestions": [],
  "session_id": "...",
  "query_id": 1235
}
```

12.2 POST /wp-json/mbr-isa/v1/feedback

Records a thumbs-up (1), thumbs-down (-1), or neutral (0) against a previously-logged query.

```
{
  "query_id": 1234,
  "feedback": 1
}
```

Feedback is only accepted within one hour of the original query. Older queries return a 410 Gone response to discourage scripted bulk-feedback.

12.3 Error responses

Code	Error	Meaning
400	mbr_isa_invalid_query_id	query_id missing or not a positive integer
400	mbr_isa_invalid_feedback	feedback value not in {-1, 0, 1}
404	mbr_isa_not_found	query_id does not exist in the log
410	mbr_isa_feedback_expired	More than an hour has passed since the query
429	mbr_isa_rate_limited	Per-IP rate limit exceeded
500	mbr_isa_db_error	Database write failed when storing feedback

13. Privacy, data, and rate limiting

13.1 What the plugin stores

- **Post content, as an inverted index.** Stored across the `mbrisa_terms`, `mbrisa_documents`, and `mbrisa_postings` tables. This is derived from content you have already published.
- **Query log.** Each query submitted through the widget is logged in `mbrisa_queries` with the query text, matched intent (if any), top score, result count, optional feedback value, session ID, created-at timestamp, and a hashed version of the requester's IP address.
- **No personal identifiers.** The plugin does not record user accounts, cookies, emails, or raw IPs.

13.2 IP hashing

Where an IP address is needed — for rate limiting and for a loose approximation of unique askers in the query log — it is immediately hashed with SHA-256 and your site's WordPress salt. The raw IP is never stored. The hashed value is not reversible without access to your site's salt.

13.3 No outbound traffic

The plugin makes no outbound HTTP requests during normal operation. Every query is answered from data on your own server. This is a core design constraint, not an optional setting.

13.4 Rate limiting

Both REST endpoints are protected by a per-IP-hash sliding window stored in WordPress transients. Defaults:

Endpoint	Default limit	Configurable via
/mbr-isa/v1/ask	30 requests / minute	rate_limit_per_min in settings
/mbr-isa/v1/feedback	20 requests / minute	Hardcoded for this release

When the limit is hit the endpoint returns 429 Too Many Requests and the widget shows a "you are asking too quickly" message.

14. Troubleshooting

The widget does not appear on the site

- Check that **Floating widget** is enabled under Widget Settings.
- Confirm your active theme calls `wp_footer()`. If it does not, the floating widget cannot render — use the shortcode instead.
- Open your browser's developer console and look for JavaScript errors from `chat-widget.js`.
- Confirm `window.mbrAisa` is defined with a `restUrl`. If not, something has stripped the localized script tag.

Widget shows "Something went wrong" on every query (401 from REST)

This is almost always caused by a security or performance plugin that disables the WordPress REST API for non-admin or logged-out users. Because the chat widget runs on the front end where most visitors are not logged in, a blanket REST blocker will catch every request.

- Check your dev tools Network tab. If the POST to `/wp-json/mbr-isa/v1/ask` returns 401 Unauthorized, this is the cause.
- If you have a REST-hardening plugin, look for an "allowed namespaces" or "REST whitelist" setting and add `mbr-isa/v1` to it. (MBR WP Performance v1.9.3 and later have this setting under *Core* → *Allowed REST Namespaces*.)
- If your plugin doesn't offer a whitelist, switch its REST mode to "Default (Enabled)" — that's the blunt fix and is fine if you trust the plugins on your site.

Queries return no results even though the content exists

- Run a full reindex. This is the most common cause on a fresh install.
- Check Index Status shows a non-zero document count.
- Use the Tokeniser Tester on both your query and the page title — see which tokens actually come out.
- If the query uses different vocabulary to your content, add a synonym group via § 9.
- Confirm the post type is listed in `enabled_post_types`.

An intent is firing when it should not

- Triggers are substrings, so a trigger like *help* will match "I need help finding the shop". Either remove the trigger or tighten it with a regex prefix (`re:\bhelp\b`).
- Use the Chat Tester to see which intent ID is firing.
- Consider toggling the **Disabled** checkbox on the intent rather than deleting it — that way you can re-enable later without re-typing the triggers.

Synonyms admin shows "websit" instead of "website" — is something broken?

No, that's the Porter stemmer doing its job. The search index uses the stemmed form internally so that "websites" matches "website" matches "webpages". The admin test panel shows the bracketed original word alongside (e.g. `websit (website)`) precisely so it doesn't read as a typo. Visitors never see stemmed forms anywhere.

The widget says "too many requests"

- You are hitting the per-IP rate limit. Wait a minute and try again.
- If you are testing programmatically, raise `rate_limit_per_min` in the plugin settings option.

Reindex takes a long time or times out

- The reindex is synchronous and batches 50 posts at a time. For very large sites (thousands of posts) it may exceed PHP's `max_execution_time`.
- Temporarily raise `max_execution_time` in `php.ini`, or run the reindex from a CLI environment where time limits do not apply.

Glass effect doesn't show on the front end

- Confirm the toggle is on in **Tools** → **MBR ISA Appearance**.
- Glass requires `backdrop-filter` support in the visitor's browser. Modern Chrome, Edge, Firefox, and Safari all support it; very old browsers fall back to a partially transparent panel without blur.
- Glass is most visible over rich backgrounds (hero images, gradients). On a plain white page it is subtle by design.

A database table is missing after activation

- Deactivate and reactivate the plugin to re-run the activator.
- Check that the WordPress database user has `CREATE TABLE` permission on your database.

15. Uninstalling and keeping data

Deactivating the plugin leaves all data intact — the tables, options, and widget settings are preserved so you can reactivate later without losing anything. Deleting the plugin (via **Plugins** → **Delete**) by default drops all four custom tables, all plugin options, and all transients.

15.1 Keeping data across a reinstall

If you plan to delete the plugin and reinstall later, set the `mbr_isa_keep_data_on_uninstall` option to a truthy value before deletion. With that option set, the uninstall routine skips all cleanup and leaves your index, query log, intents, synonyms, and theme settings untouched.

```
update_option( 'mbr_isa_keep_data_on_uninstall', 1 );
```

16. Technical reference

16.1 Plugin settings option

All widget, engine, and theme configuration lives in one option: `mbr_isa_settings`. The structure and defaults:

Key	Default	Purpose
<code>enabled_post_types</code>	<code>['post', 'page']</code>	Which post types are indexed.
<code>bm25_k1</code>	1.2	BM25 term-frequency saturation constant.
<code>bm25_b</code>	0.75	BM25 length-normalisation constant.
<code>field_weight_title</code>	3.0	Score multiplier for title-field matches.
<code>field_weight_excerpt</code>	1.5	Score multiplier for excerpt matches.
<code>field_weight_body</code>	1.0	Score multiplier for content matches.
<code>widget_enabled</code>	false	Show floating bubble site-wide.
<code>widget_position</code>	'bottom-right'	'bottom-right' or 'bottom-left'.
<code>widget_title</code>	(empty)	Overrides panel title if set.
<code>widget_greeting</code>	(empty)	Overrides greeting message if set.
<code>widget_placeholder</code>	(empty)	Overrides input placeholder if set.
<code>theme_preset</code>	'mocha'	One of: mocha, slate-light, ocean, sunset, forest. (v0.6.0+)
<code>theme_glass</code>	false	Enable glassmorphism effect on the widget. (v0.6.0+)
<code>log_queries</code>	true	Whether to log queries.
<code>rate_limit_per_min</code>	30	Max /ask requests per minute, per IP.

16.2 Other options

Option	Purpose
mbr_isa_version	Installed plugin version string.
mbr_isa_db_version	Schema version for future migrations.
mbr_isa_intents	Array of intent definitions (defaults used if empty). Includes 'disabled' bool from v0.4.0.
mbr_isa_synonyms	Array of synonym groups (defaults used if empty).
mbr_isa_index_status	Cached counts shown on the diagnostic page.
mbr_isa_keep_data_on_uninstall	If truthy, skip all cleanup on delete.

16.3 Intent data shape

Each entry in `mbr_isa_intents` is an associative array:

```
[
  'id'      => 'contact',
  'label'   => 'Contact',
  'triggers' => [ 'contact', 'email', 'get in touch' ],
  'response' => 'You can reach me at <a href="...">hello@example.com</a>.',
  'confidence' => 1.0,
  'disabled' => false,    // v0.4.0+, optional, defaults to false
]
```

16.4 File layout

```

mbr-intelligent-site-assistant/
|-- mbr-intelligent-site-assistant.php    Main plugin bootstrap
|-- readme.txt
|-- uninstall.php                        Full-cleanup routine
|-- assets/
|   |-- css/chat-widget.css              Widget styling + theme presets + glass
|   |-- js/chat-widget.js                Widget controller
|-- data/
|   |-- stopwords-en.php                 English stopword list
|-- includes/
|   |-- class-mbr-isa-activator.php       Creates tables, seeds options
|   |-- class-mbr-isa-deactivator.php     Deactivation hooks
|   |-- class-mbr-isa-tokeniser.php       Clean / split / stem
|   |-- class-mbr-isa-bm25.php            BM25 scoring
|   |-- class-mbr-isa-indexer.php         Index build, search, field weights
|   |-- class-mbr-isa-synonyms.php        Synonym expansion
|   |-- class-mbr-isa-intents.php         Intent matching
|   |-- class-mbr-isa-responder.php       Confidence, messages, snippets
|   |-- class-mbr-isa-query-handler.php   Orchestrates the pipeline
|   |-- class-mbr-isa-rate-limiter.php    Per-IP throttling
|   |-- class-mbr-isa-rest.php            REST route registration
|   |-- class-mbr-isa-frontend.php        Shortcode + floating widget render
|   |-- class-mbr-isa-admin-intents.php   Admin UI for intents (v0.4.0+)
|   |-- class-mbr-isa-admin-synonyms.php  Admin UI for synonyms (v0.5.0+)
|   |-- class-mbr-isa-admin-theme.php     Admin UI for appearance (v0.6.0+)
|   |-- class-mbr-isa.php                 Singleton orchestrator + diagnostic page

```

16.5 JavaScript integration

The widget exposes a configuration object at `window.mbrAisa`, populated by `wp_localize_script`:

```

window.mbrAisa = {
  restUrl: 'https://example.com/wp-json/mbr-isa/v1/ask',
  feedbackUrl: 'https://example.com/wp-json/mbr-isa/v1/feedback',
  nonce: '...',
  strings: { sending: 'Searching...', networkErr: '...', ... }
};

```

For manual re-initialisation (for example, after an AJAX page swap in a headless theme) the widget exposes `window.MbrAisaChat.init()`.

Note. The JavaScript-side identifiers still use the camelCase `mbrAisa` form, retained from an earlier version of the plugin for backward compatibility with any custom front-end integration code. All PHP-side identifiers are `mbr_isa`.

17. Support and version information

Plugin version	0.6.0
Author	Robert Palmer
Distribution	littlewebshack.com
Licence	GPL-2.0-or-later
Text domain	mbr-isa
Requires	WordPress 5.8+, PHP 7.4+

This plugin is free and open source. If it has saved you time, a donation at buymeacoffee.com/robertpalmer is very welcome.

For bug reports, feature suggestions, or support questions, please get in touch through littlewebshack.com or via the author's contact channels.